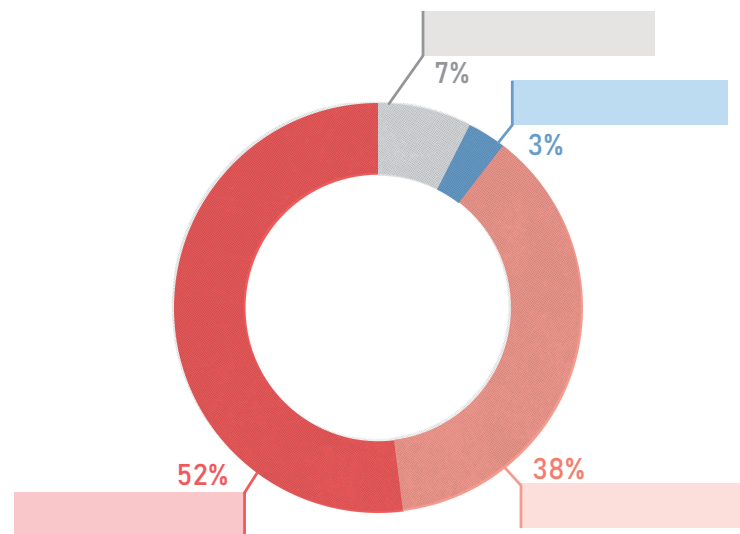


In-Class Exercise Sheet 1

These questions are based on (1) the Nature article on “Reproducibility Crisis”¹ and (2) the weblink to the ACM terms on “Artifact Review and Badging”².

1. 1,576 researchers were asked if there is a reproducibility crisis. This was their answer:



2. Are we talking about repeatability/reproducibility/replicability?

- (a) Scientists frequently perform the experiments of other scientists. Why is this important in science?
- ☐ Scientists often ask each other for help in performing experiments.
 - ☐ They do this to practice their own technique.
 - ☐ This helps confirm and verify the results.
 - ☐ This is an effective way to ensure that a laboratory is in optimal condition.
- (b) Alice has been reading about another scientist's research in a physics journal. She plans to do these experiments herself, by reconstructing the setup described in the journal article, and to see whether she gets the same result.
- ☐ Repetition
 - ☐ Reproduction
 - ☐ Replication
- (c) Bob goes to the river Inn every week for 4 weeks and collects 5 samples of water to test the pH.
- ☐ Repetition
 - ☐ Reproduction

¹<https://doi.org/10.1038/d41586-019-00067-3>

²<https://www.acm.org/publications/policies/artifact-review-and-badging-current>

- ☐ Replication
- (d) An environmental group contacts Nina to obtain her artifacts (software, input data, equipment, ...) so that they can check her simulation results on climate change.
- ☐ Repetition
☐ Reproduction
☐ Replication
- (e) Charlie was mixing a batch of glue and borax solution to make silly putty and discovered a new substance that glows in the dark. He asks Alice to visit him in his lab, use his specific amounts of ingredients and the steps of his procedure. He asked Alice to try the experiment to see if she gets the same results.
- ☐ Repetition
☐ Reproduction
☐ Replication
- (f) Dave baked chocolate chip cookies that wound up looking like muffins. He asked Eve to follow the same directions and see if, baking in her own kitchen, her cookies also look like muffins.
- ☐ Repetition
☐ Reproduction
☐ Replication
- (g) Fay has taught her hamster Mini Bear to run a maze. She thinks that the hamster will go faster if she puts its favorite treat in the end. She has the hamster run the maze ten times with her favorite treat and ten times with a regular food pellet. She uses a stopwatch to measure how long it takes for the hamster to get to the reward.
- ☐ Repetition
☐ Reproduction
☐ Replication
- (h) George is testing golf balls. He drops three different brands of balls from 40 cm and measures how high they bounce. He drops each ball 10 times.
- ☐ Repetition
☐ Reproduction
☐ Replication
- (i) Harry was given a set of 5 metal cubes. He calculated the densities of each to determine the type of metal in each. The densities were really close to each other, so he asked his friend Joy to come over and do the same experiment, using his equipment, to see if she gets the same results.
- ☐ Repetition
☐ Reproduction
☐ Replication

The above questions are adapted from the Quizizz website.

3. ACM defines Artifact Availability as follows:

Author-created artifacts relevant to this paper have been placed on a publically accessible archival repository. A DOI or link to this repository along with a unique identifier for the object is provided.

How would you ensure artifact availability for the research artifacts produced in your Masters thesis research?

4. Remember your own Bachelors thesis project.

- (a) What was your research question? Formulate it as a one-sentence hypothesis.

- (b) How did you confirm/reject your hypothesis? Did you write a program, did you run software-based experiments?

- (c) What were the research artifacts that you produced in your thesis? List them.

- (d) If you were given 24 hours, would you be able to repeat your experiments from back then? This includes getting code to run and re-generating all the tables or charts that you reported in your thesis. Describe any challenges that you expect.

- (e) If you could turn back time to the day when you submitted your Bachelor thesis, how would you ensure that you are enabled to repeat your experiment from back then *today*?

These questions were inspired by the worksheets in the book “Forschungsdatenmanagement in der Informatik” by Katarzyna Biernacka and Sandra Schulz, DOI <https://doi.org/10.30819/5490>.